Thread-Safe Hash Index

Parallel Index Construction

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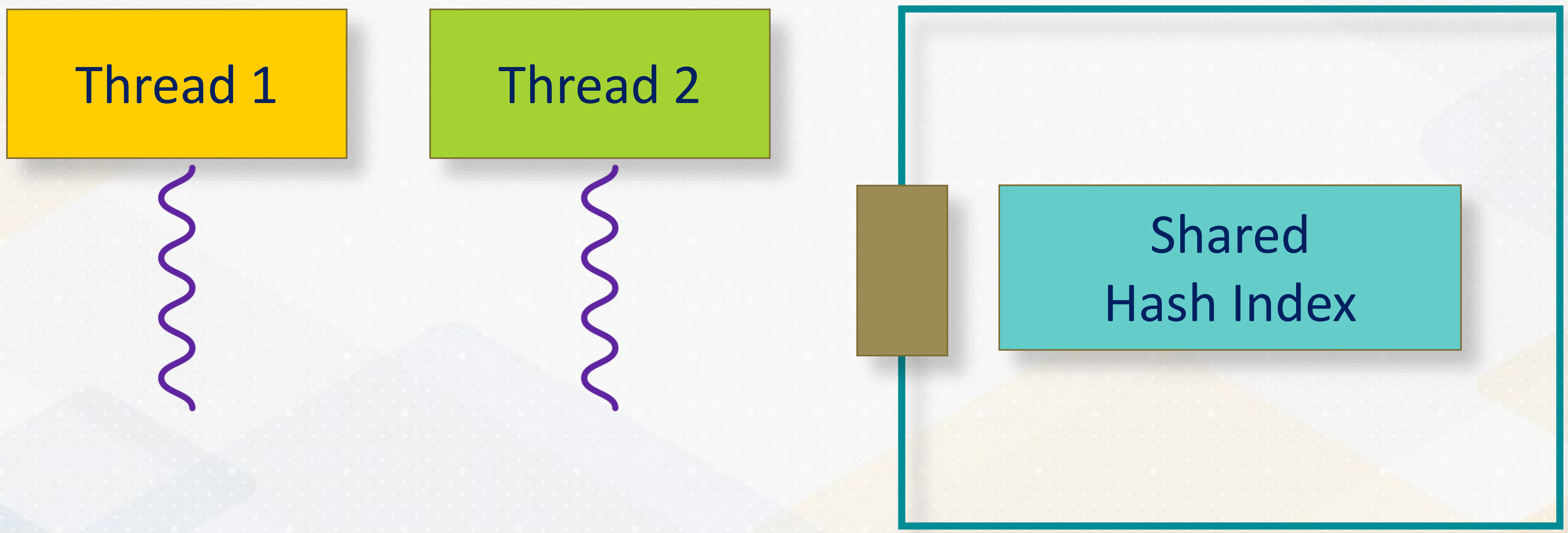
Thread Safety

Parallel Index Construction

With multi-core CPUs, **parallelizing index construction** offers a significant performance boost by **distributing** the workload across **multiple threads**.

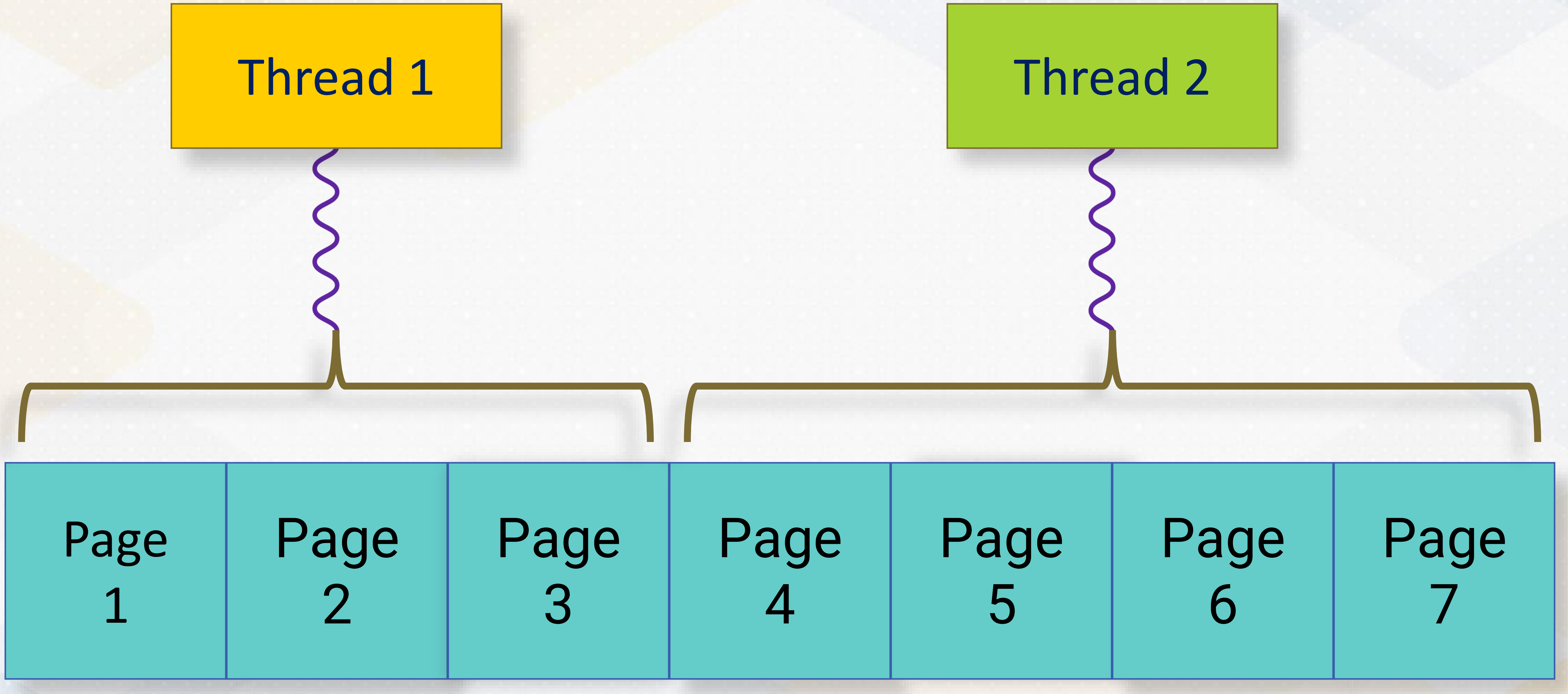


Page Assignment

Divide the total number of pages (num_pages) by the number of available threads (num_threads), assign each thread a specific range of pages to process.

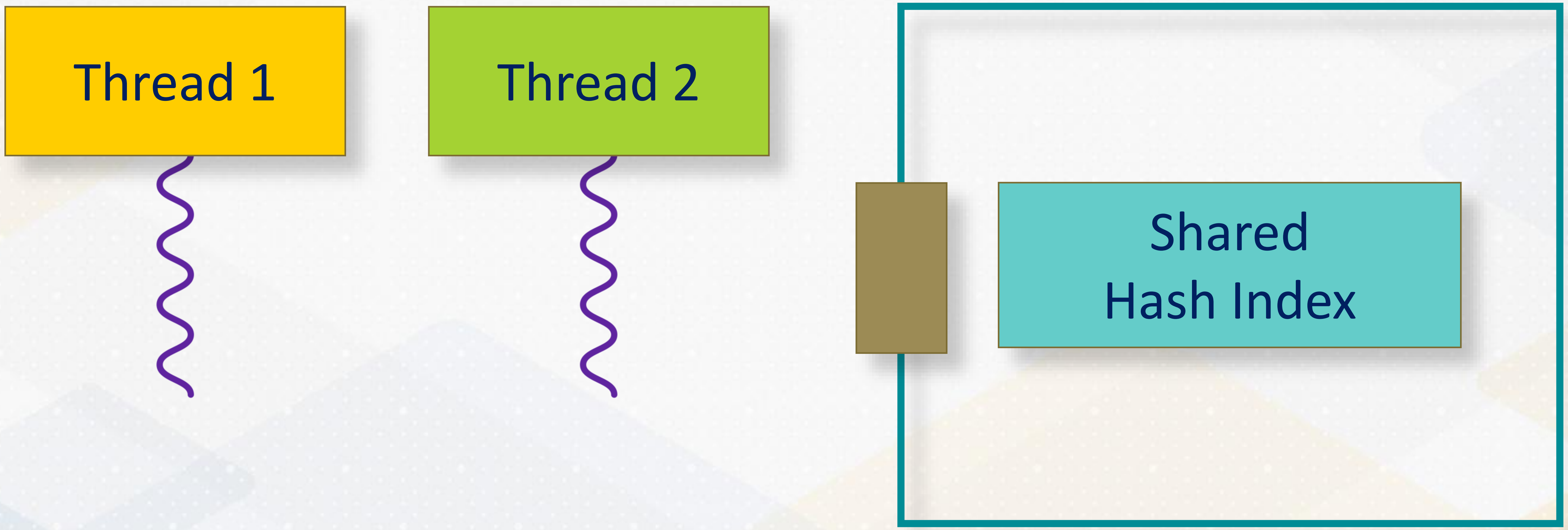
```
void parallelProcessPages(size_t num_threads = 5) {
  auto num_pages = buffer_manager.getNumPages();
  size_t pages_per_thread = num_pages / num_threads;
  std::vector<std::thread> threads;
  for (size_t i = 0; i < num_threads; i++) {
    size_t start_page = i * pages_per_thread;
    size_t end_page =…; // Last thread gets any remaining pages
    threads.emplace_back(&BuzzDB::processPageRange, this, start_page, end_page);
  }
  …
}
```

Page Assignment



Thread Safety

With **parallel index construction**, the challenge is that **concurrent operations** on the index by multiple threads may lead to **inconsistent state**.



Thread Safety

Thread 1: Insert (15, Fig) **Thread 2**: Insert (25, Pear)

0	1	2	3	4	5	6	7	8	9

HASH(KEY) = KEY % 10

Thread Safety

Thread 1: Insert (15, Fig) **Thread 2**: Insert (25, Pear)

0	1	2	3	4	5	6	7	8	9
					15				
					Fig				

HASH(KEY) = KEY % 10

Thread Safety

Thread 1: Insert (15, Fig) **Thread 2**: Insert (25, Pear)

0	1	2	3	4	5	6	7	8	9
					25				
					Pear				

HASH(KEY) = KEY % 10

Race Condition

	Thread 1	Thread 2
1	HASH key 15	
2	PROBE slot 5	
3		HASH key 25
4		PROBE slot 5
5	STORE key 15 in slot 5	
6		STORE key 25 in slot 5

Time

Race Condition

	Thread 1	Thread 2
1	HASH key 15	
2	PROBE slot 5	
3		HASH key 25
4		PROBE slot 5
5	STORE key 15 in slot 5	
6		STORE key 25 in slot 5

Time

Race Condition

	Thread 1	Thread 2
1	HASH key 15	
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Time

Race Condition

	Thread 1	Thread 2
1	HASH key 15	
2	PROBE slot 5	
3		HASH key 25
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5	STORE key 15 in slot 5	
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Time

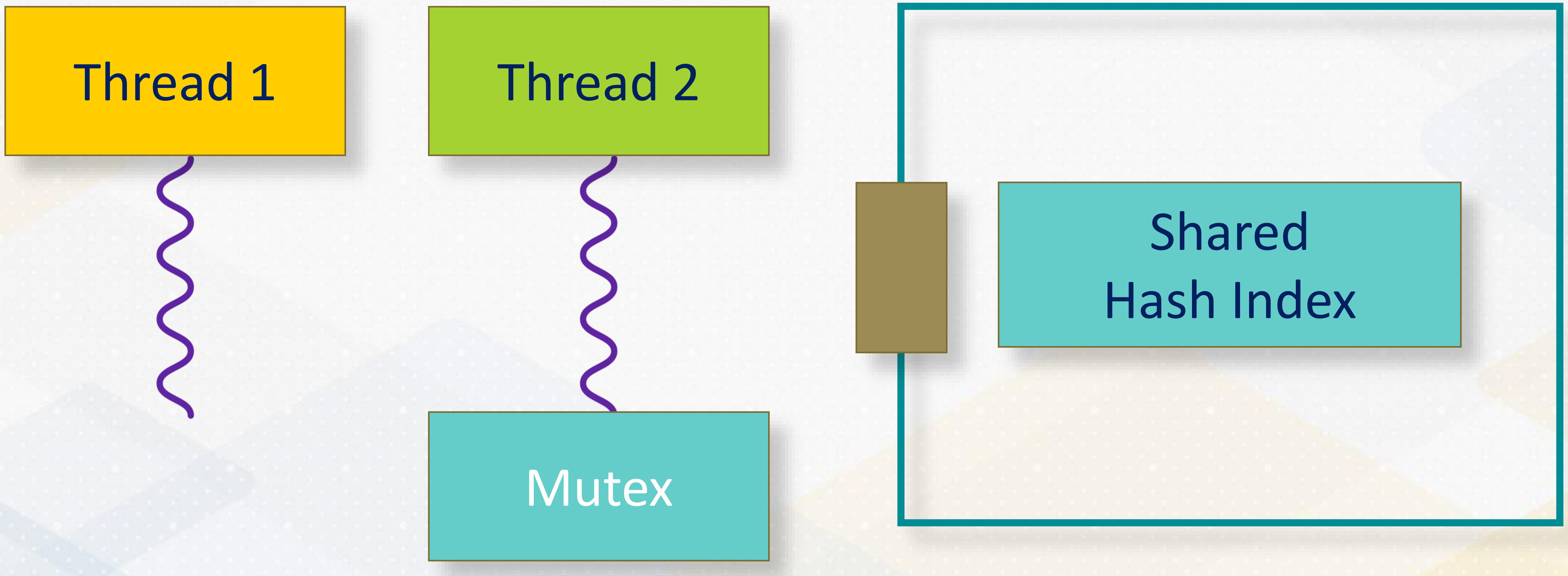
Race Condition

	Thread 1	Thread 2
1	HASH key 15	
2	PROBE slot 5	
3		HASH key 25
4		PROBE slot 5
5	STORE key 15 in slot 5	
6		STORE key 25 in slot 5

Time

std::mutex

Mutex "serializes" access to the shared index structure.



std::lock_guard

lock_guard automatically acquires a lock on creation and releases it on destruction.

```
class HashIndex {
private:
  mutable std::mutex indexMutex; // Mutex for thread-safe access

  void insertOrUpdate(int key, int value) {
    // RAII-style mutex management
    std::lock_guard<std::mutex> guard(indexMutex);
    // Perform thread-safe update on the index
  }
};
```

Thread Safety with Mutex

	Thread 1	Thread 2
1	Lock Index Mutex	
2	HASH key 15	
3	PROBE slot 5	WAIT on Index Mutex
4	**STORE key 15 in slot 5**	WAIT on Index Mutex
5	Unlock Index Mutex	WAIT on Index Mutex
6		Lock Index Mutex
7		HASH key 25
8		PROBE slot 5 and see key 15
9		PROBE slot 6
10		**STORE key 25 in slot 6**
11		Unlock Index Mutex

Time

Thread Safety with Mutex

Time	Thread 1	Thread 2
1	Lock Index Mutex	
2	HASH key 15	
3	PROBE slot 5	WAIT on Index Mutex
4	**STORE key 15 in slot 5**	WAIT on Index Mutex
5	Unlock Index Mutex	WAIT on Index Mutex
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8		PROBE slot 5 and see key 15
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Thread Safety with Mutex

Time	Thread 1	Thread 2
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9		PROBE slot 6
10		**STORE key 25 in slot 6**
11		Unlock Index Mutex

Thread Safety with Mutex

	Thread 1	Thread 2
1		Lock Index Mutex
2		HASH key 25
3	WAIT on Index Mutex	PROBE slot 5
4	WAIT on Index Mutex	**STORE key 25 in slot 5**
5	WAIT on Index Mutex	Unlock Index Mutex
6	Lock Index Mutex	
7	HASH key 15	
8	PROBE slot 5 and see key 25	
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Time

Thread Safety with Mutex

Time	Thread 1	Thread 2
1		Lock Index Mutex
2		HASH key 25
3	WAIT on Index Mutex	PROBE slot 5
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Thread Safety with Mutex

Time	Thread 1	Thread 2
1		Lock Index Mutex
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3	WAIT on Index Mutex	PROBE slot 5
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7	HASH key 15	
8	PROBE slot 5 and see key 25	
9	PROBE slot 6	
10	**STORE key 15 in slot 6**	
11	Unlock Index Mutex	

Order of Thread Execution

Thread 1 → Thread 2

0	1	2	3	4	5	6	7	8	9
					15	25			
					Fig	Pear			

Thread 2 → Thread 1

0	1	2	3	4	5	6	7	8	9
					25	15			
					Pear	Fig			

Parallel Index Construction

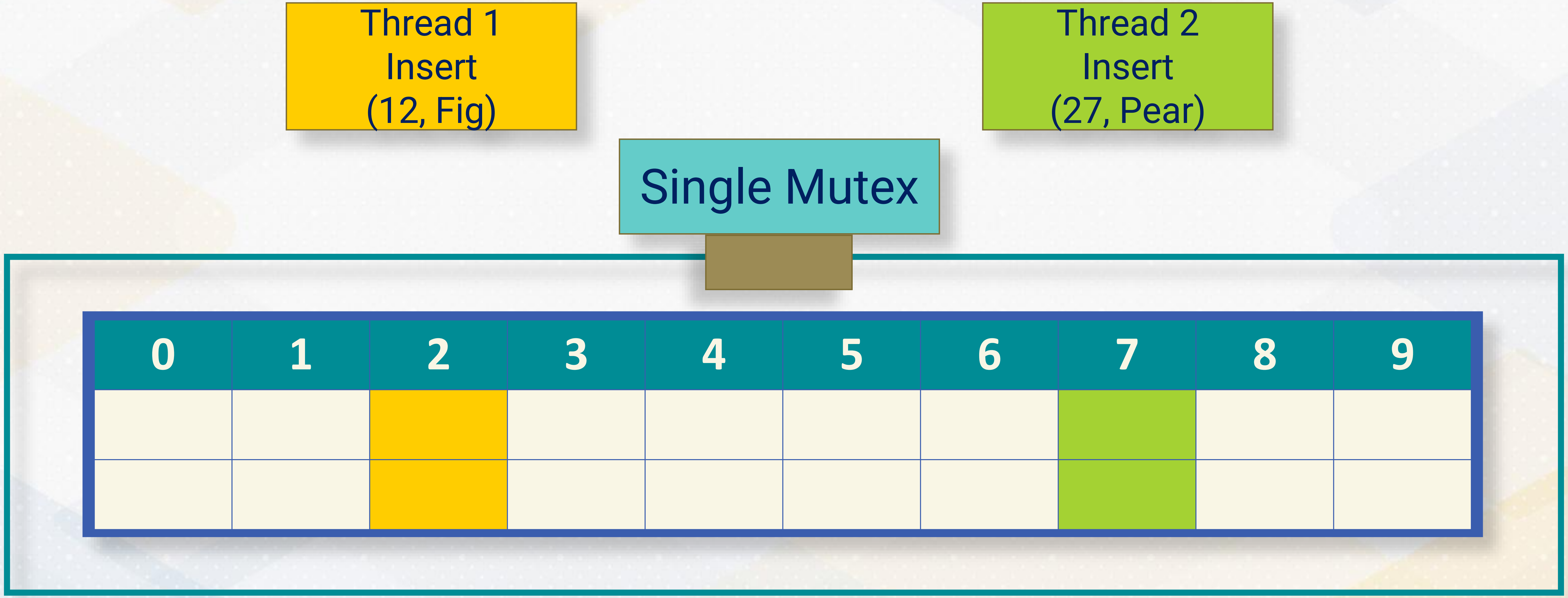
Parallel Index Construction

Thread Safety

Fine-Grained Locking

Limited Concurrency

Using a single lock for the entire hash table severely limits parallelism.



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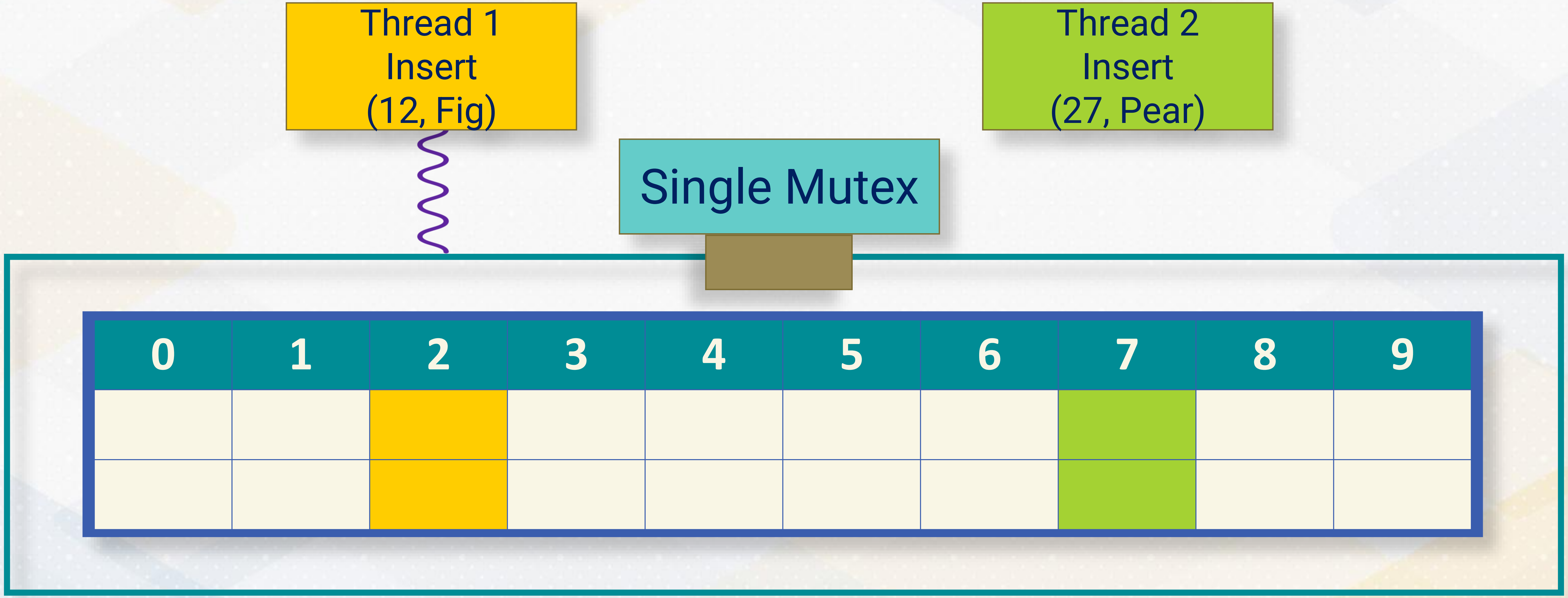

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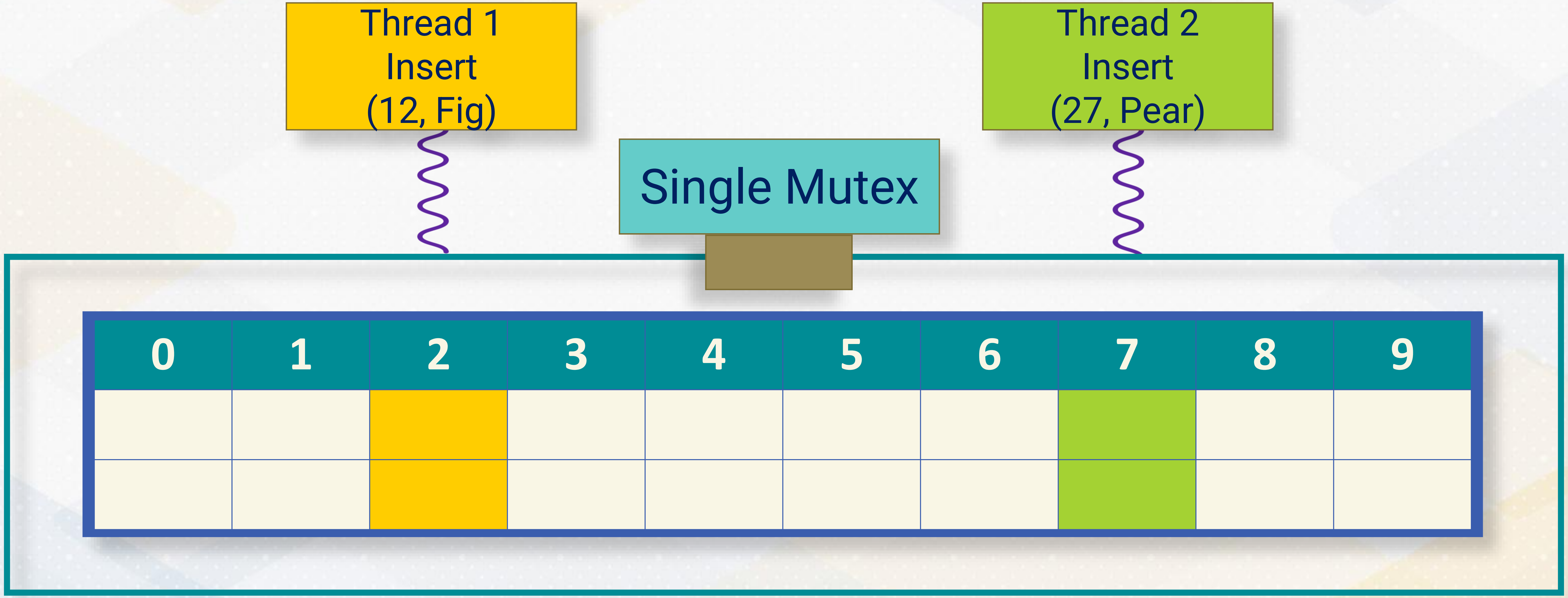
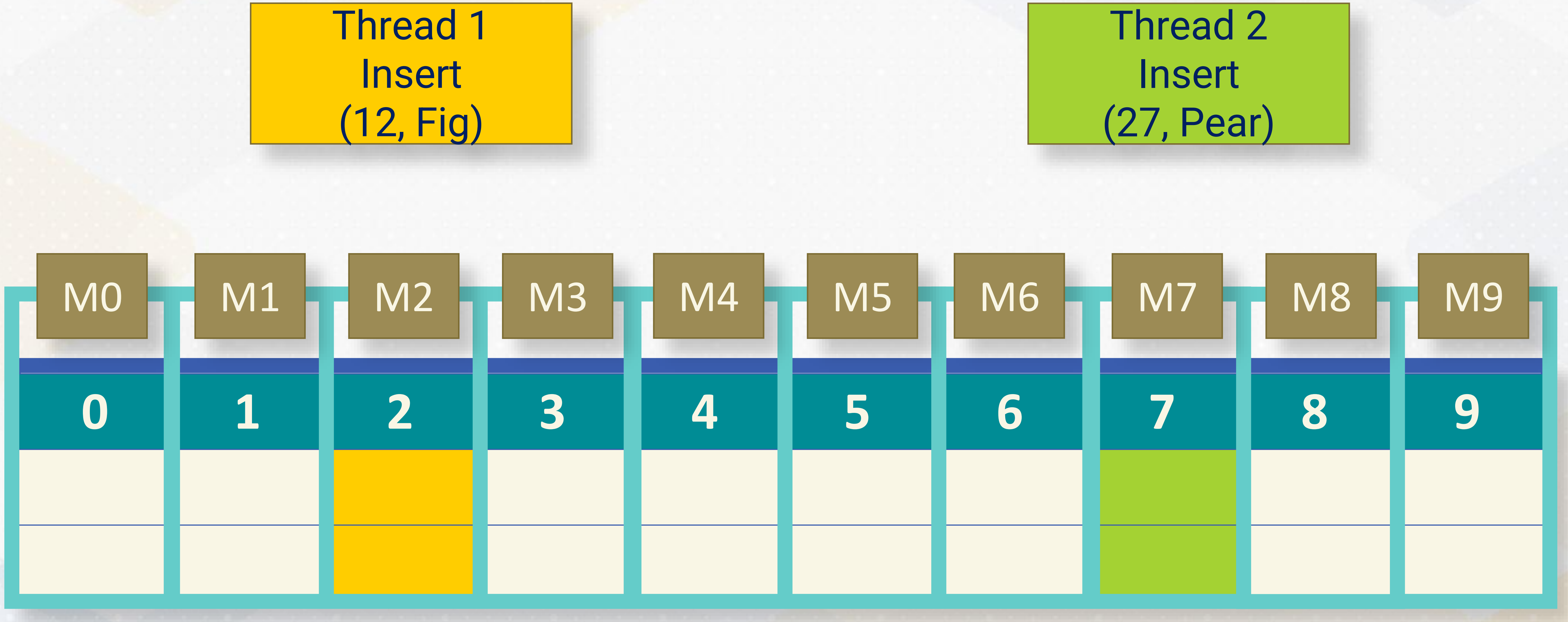


Fine-Grained Locking

Using a lock for each slot increases parallelism.



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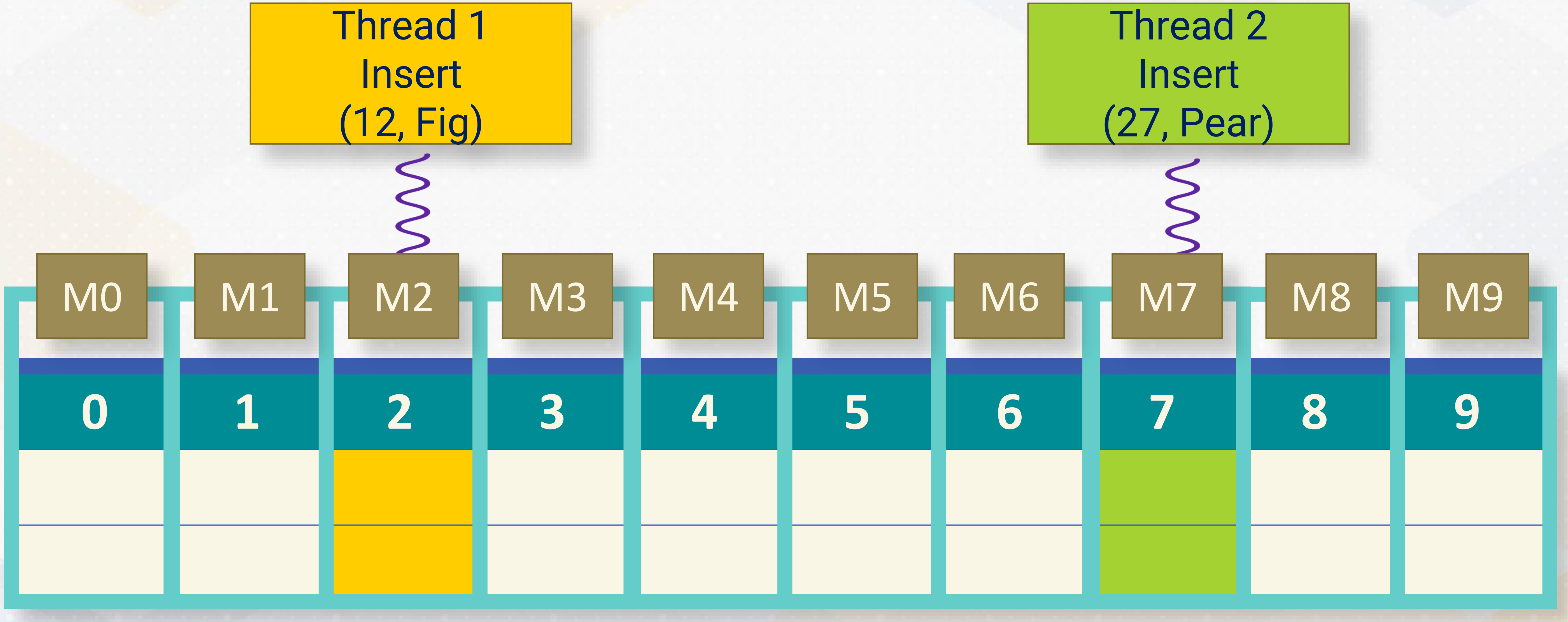


Fine-Grained Locking

Each slot in the hash table has an associated mutex.

```
std::vector<std::unique_ptr<std::mutex>> mutexes;
void insertOrUpdate(int key, int value) {
  size_t index = hashFunction(key); // Determine the slot index for the key
  do {
    // Lock only the mutex for the specific slot
    std::lock_guard<std::mutex> lock(*mutexes[index]);
    // Attempt to insert or update the slot
    if (conditions_met) {…}
  } while (not_inserted);
}
```

vector<mutex> vs vector<unique_ptr<mutex>>

Vector<element> requires element to be movable.

Mutex
NOT MOVABLE

unique_ptr<mutex>
MOVABLE

Fine-Grained Locking

By locking only the specific slot being accessed, rather than the entire hash table, fine-grained locking enables higher concurrency.

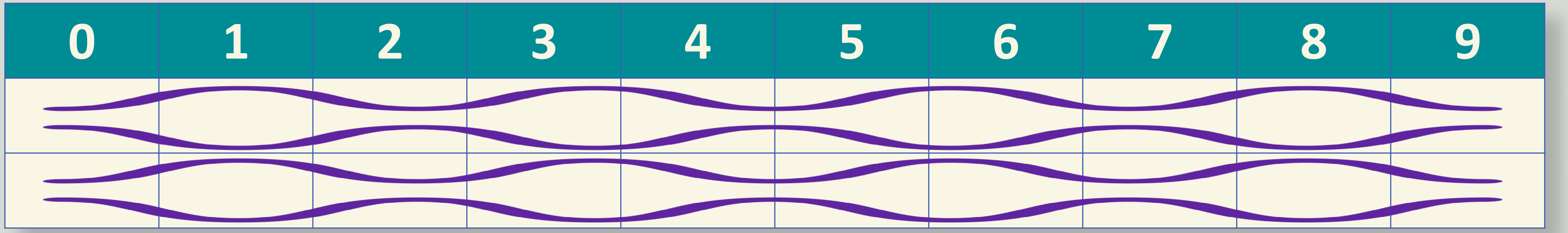
```
int getValue(int key) const {
    size_t index = hashFunction(key);
    size_t originalIndex = index;
    do {
        // Lock only the specific slot's mutex
        std::lock_guard<std::mutex> lock(*mutexes[index]);
        // Check if the key is inside the slot or not
        // Calculate next slot index
    } while (index != originalIndex);
}
```

Benefits of Fine-Grained Locking

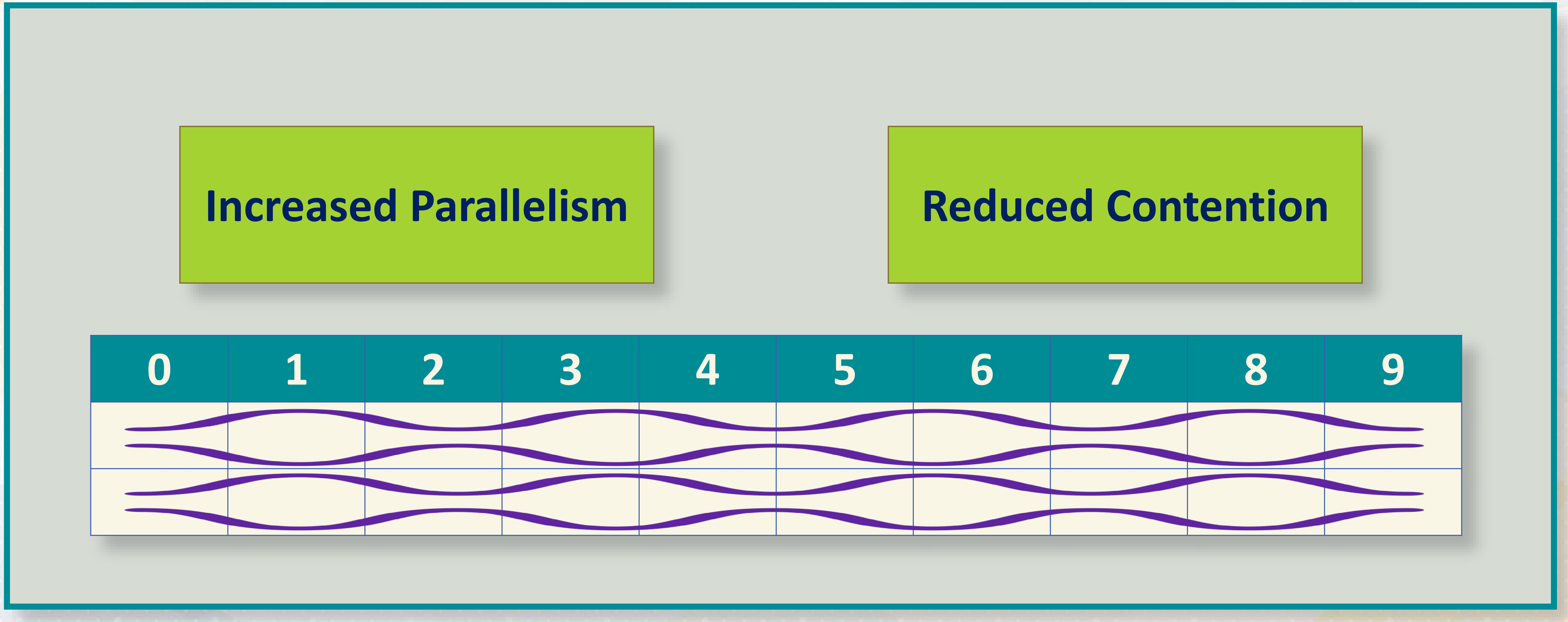
0	1	2	3	4	5	6	7	8	9

Benefits of Fine-Grained Locking

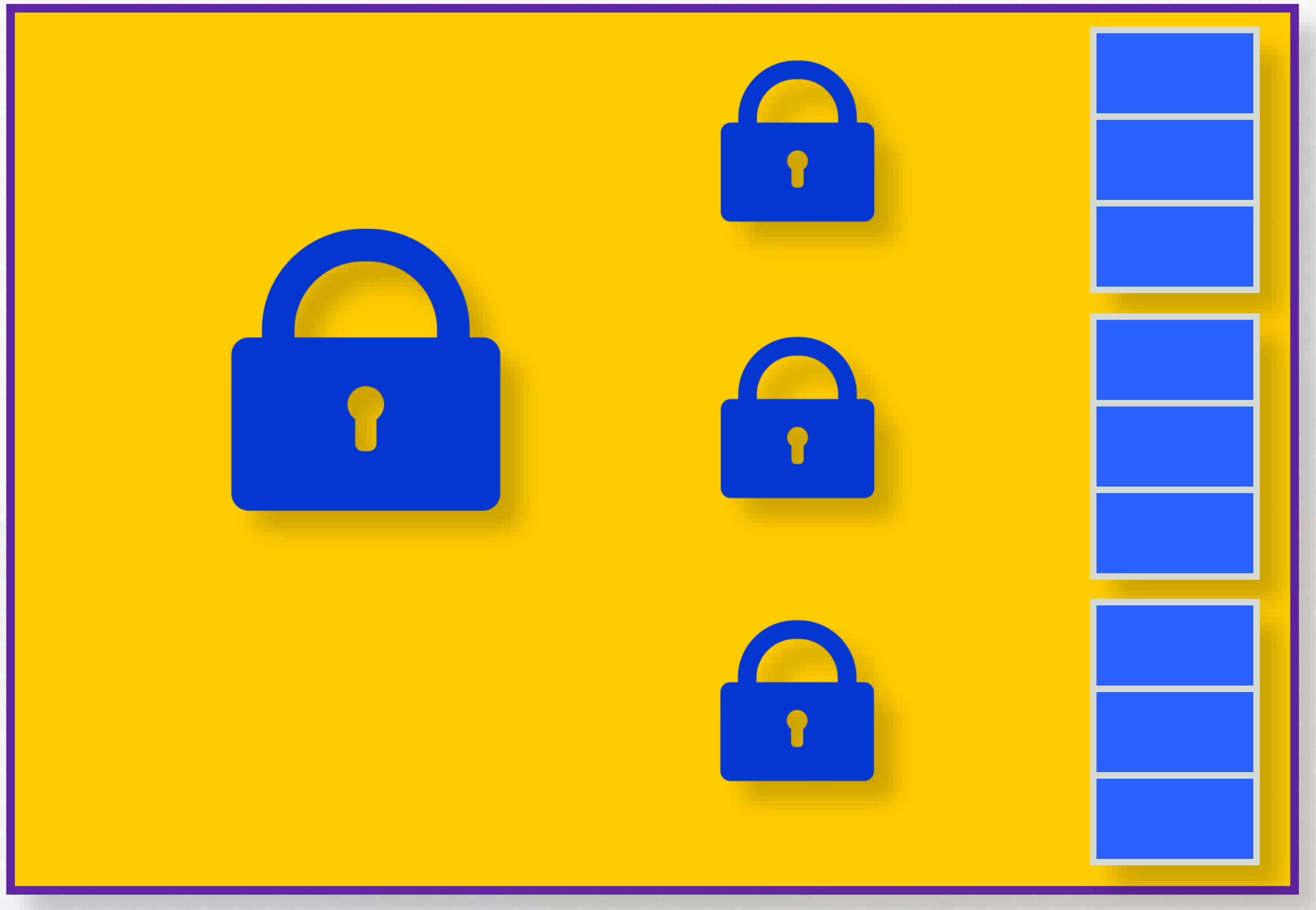
Increased Parallelism



Benefits of Fine-Grained Locking

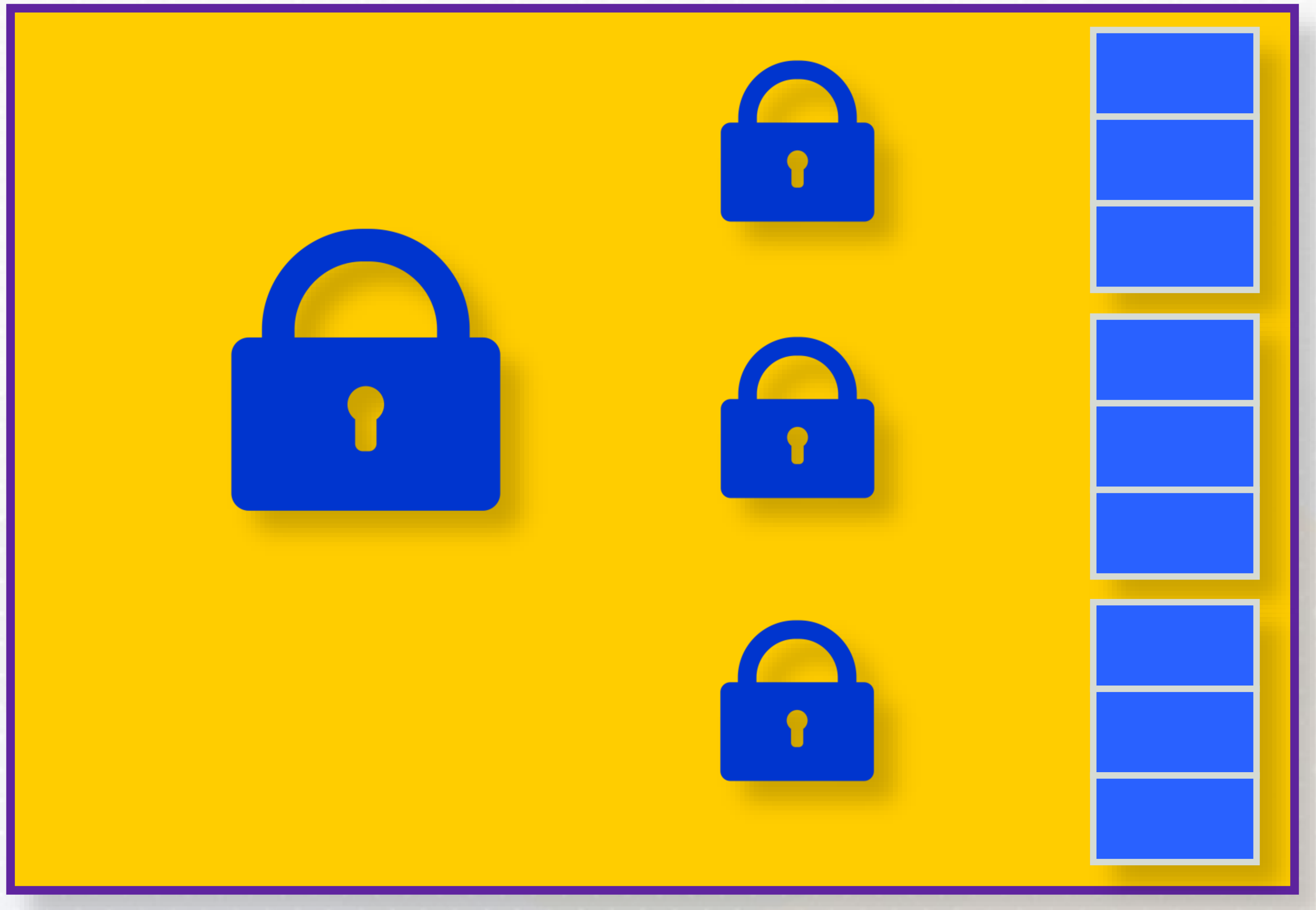


Limitations of Fine-Grained Locking



Limitations of Fine-Grained Locking

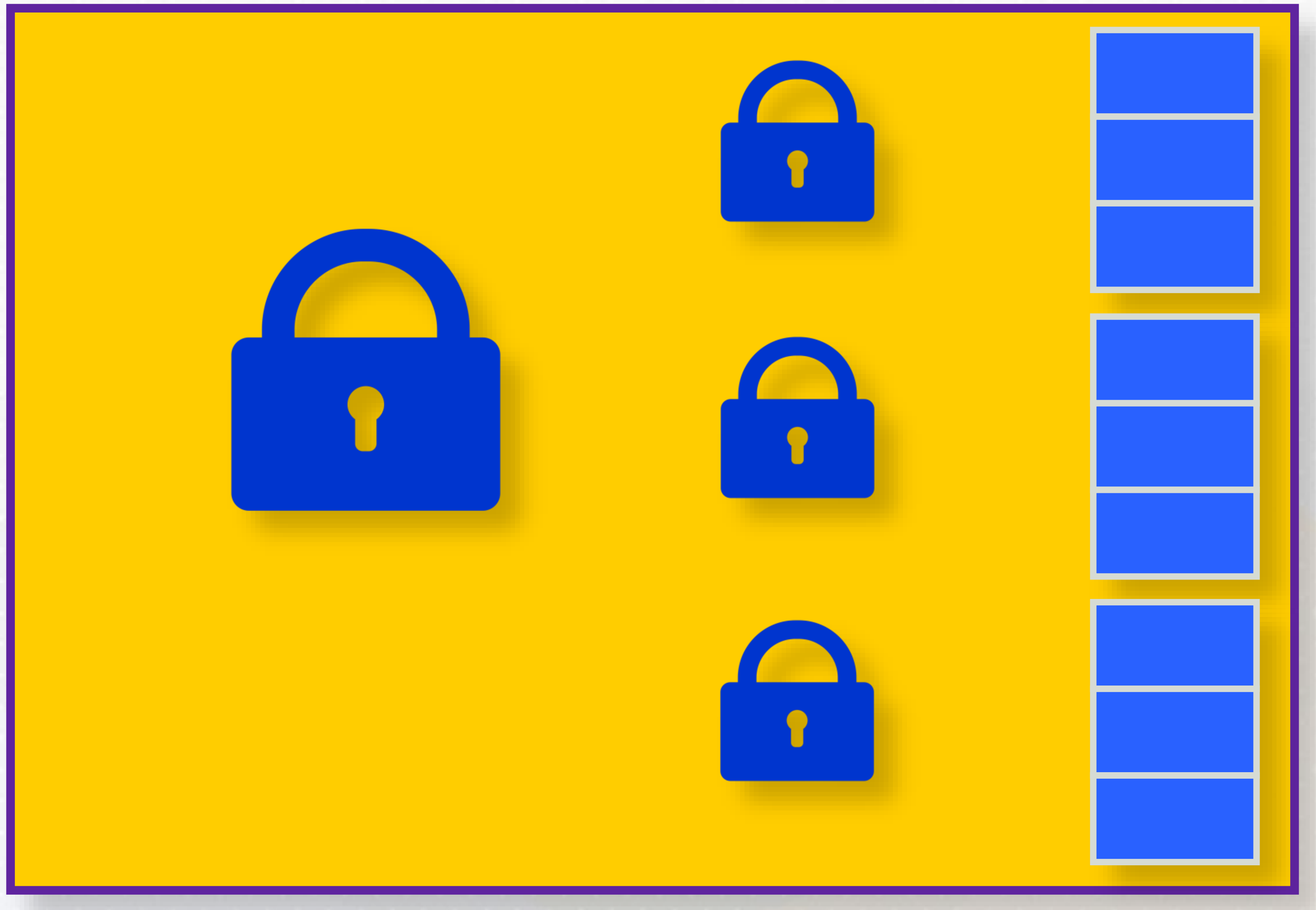
Increased Lock Acquisition/ Release Cost



Limitations of Fine-Grained Locking

Increased Lock Acquisition/ Release Cost

Increased Lock Memory Consumption



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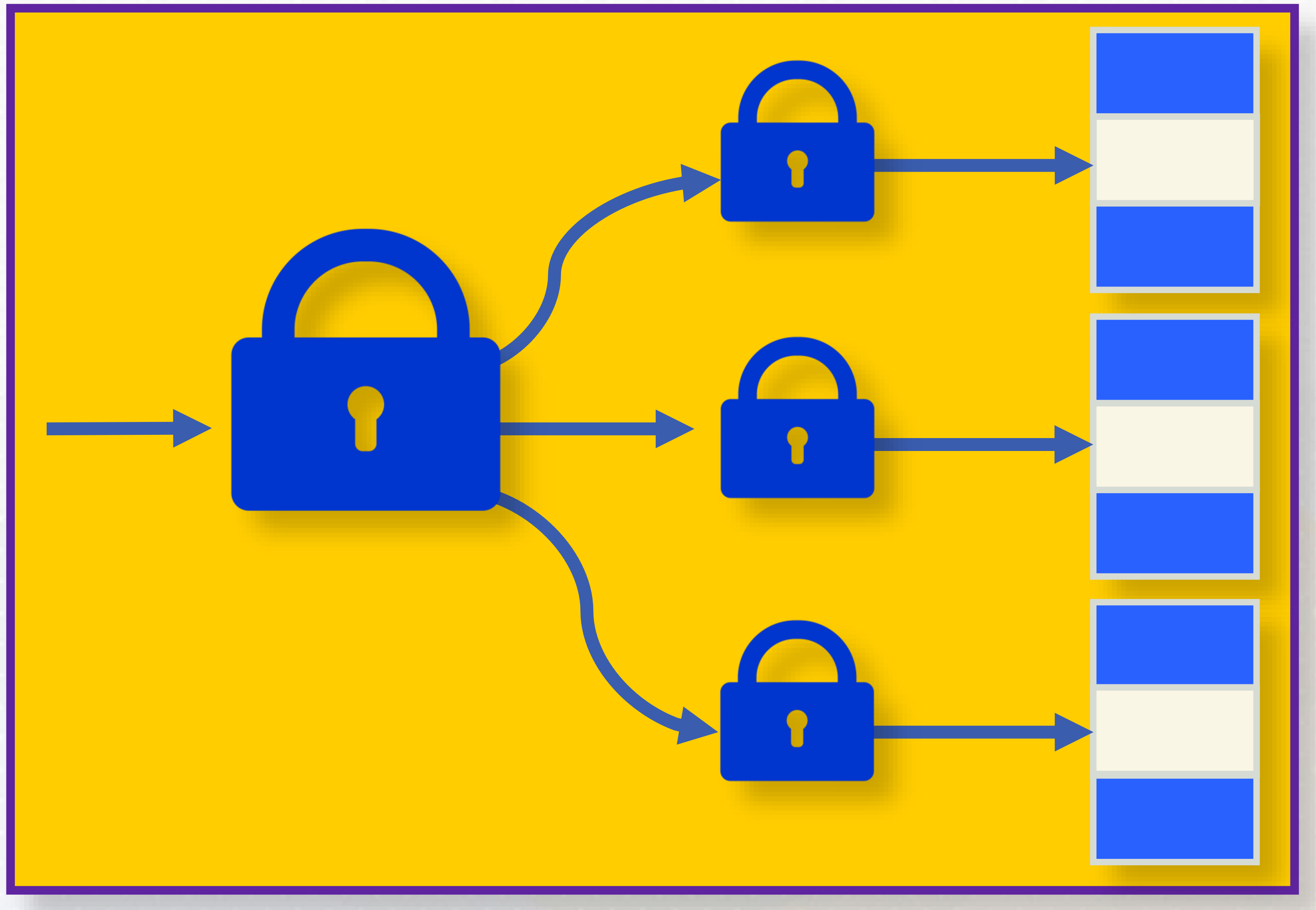
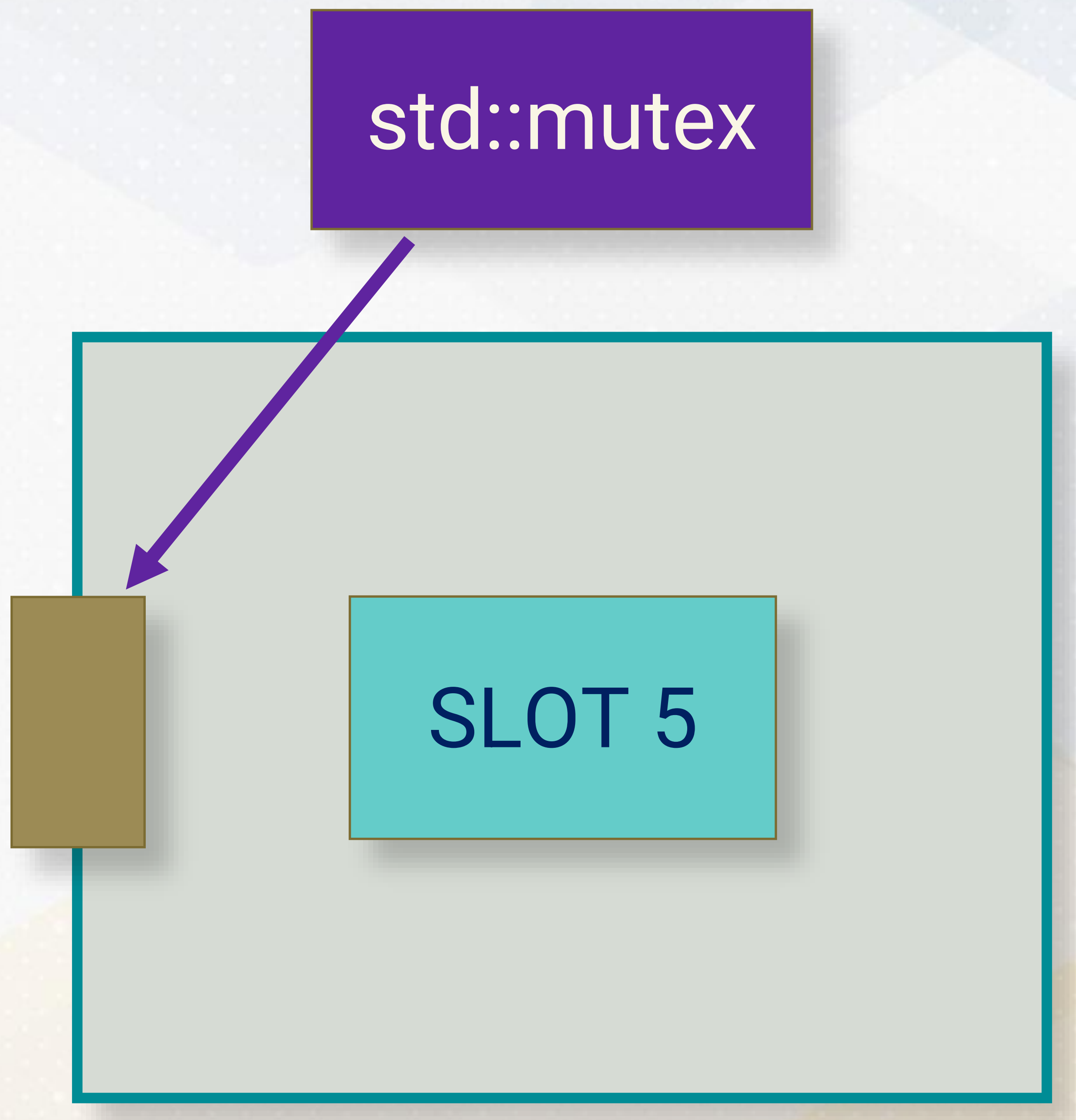


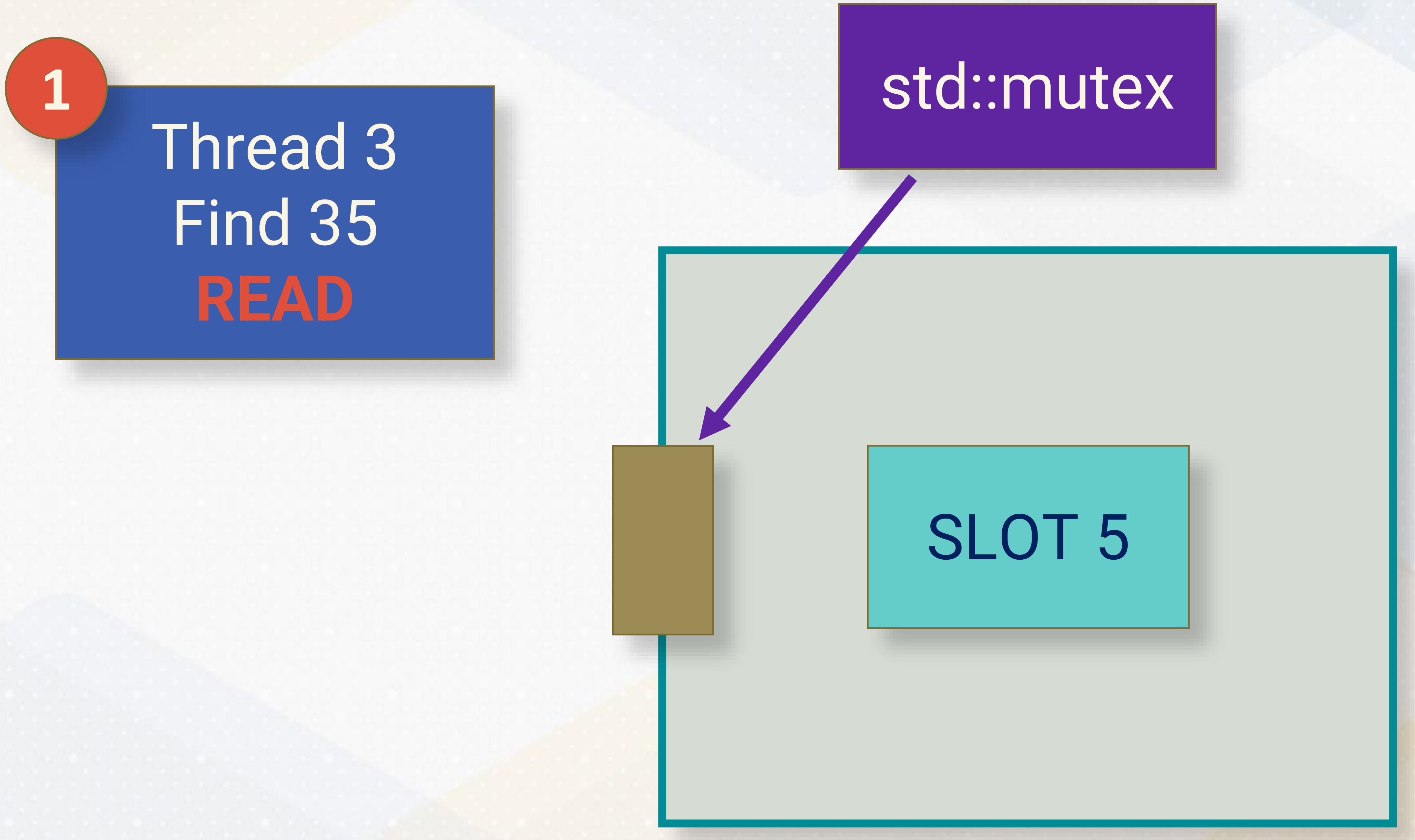
Fine-Grained Locking

Shared Mutex

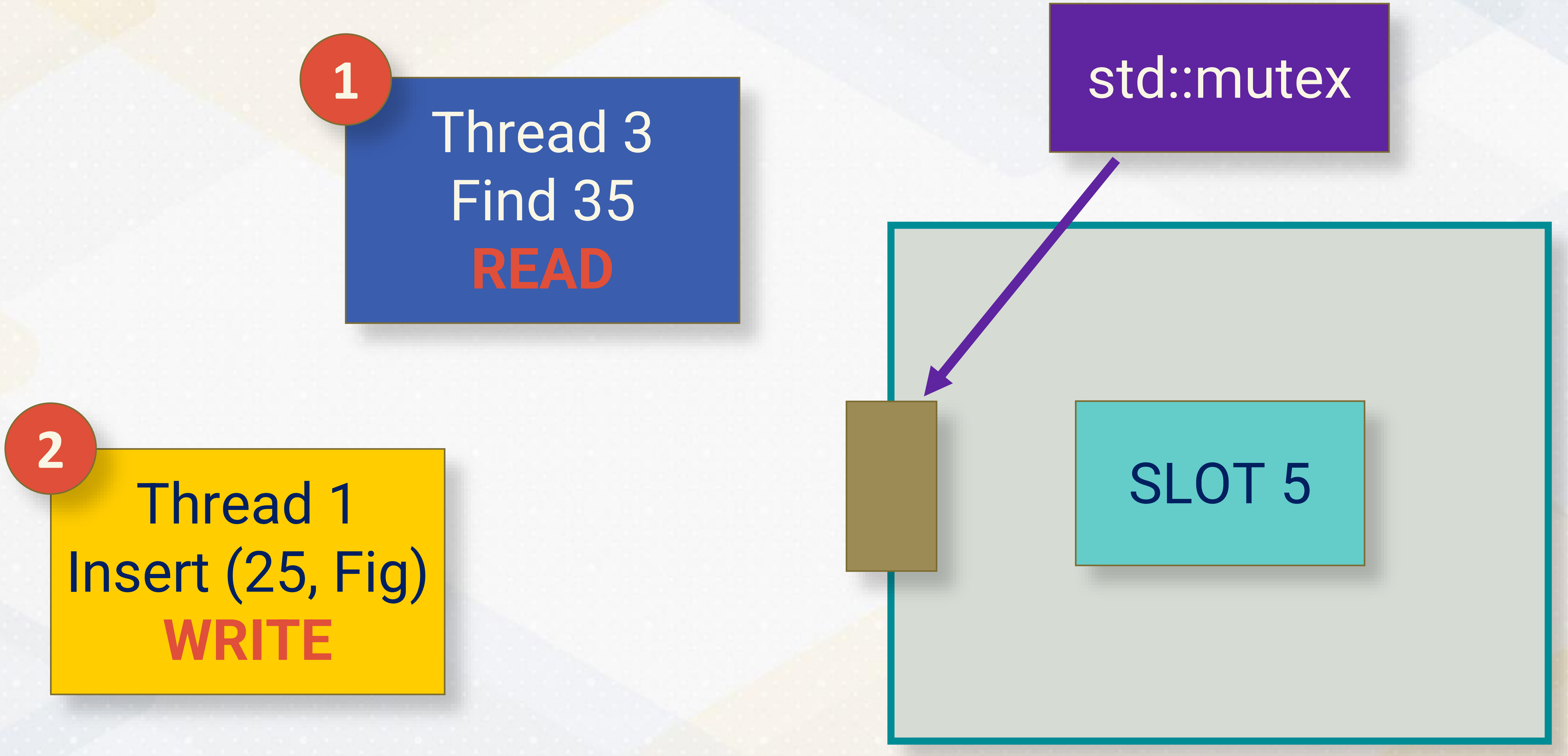
Limitations of std::mutex



Limitations of std::mutex



Limitations of std::mutex



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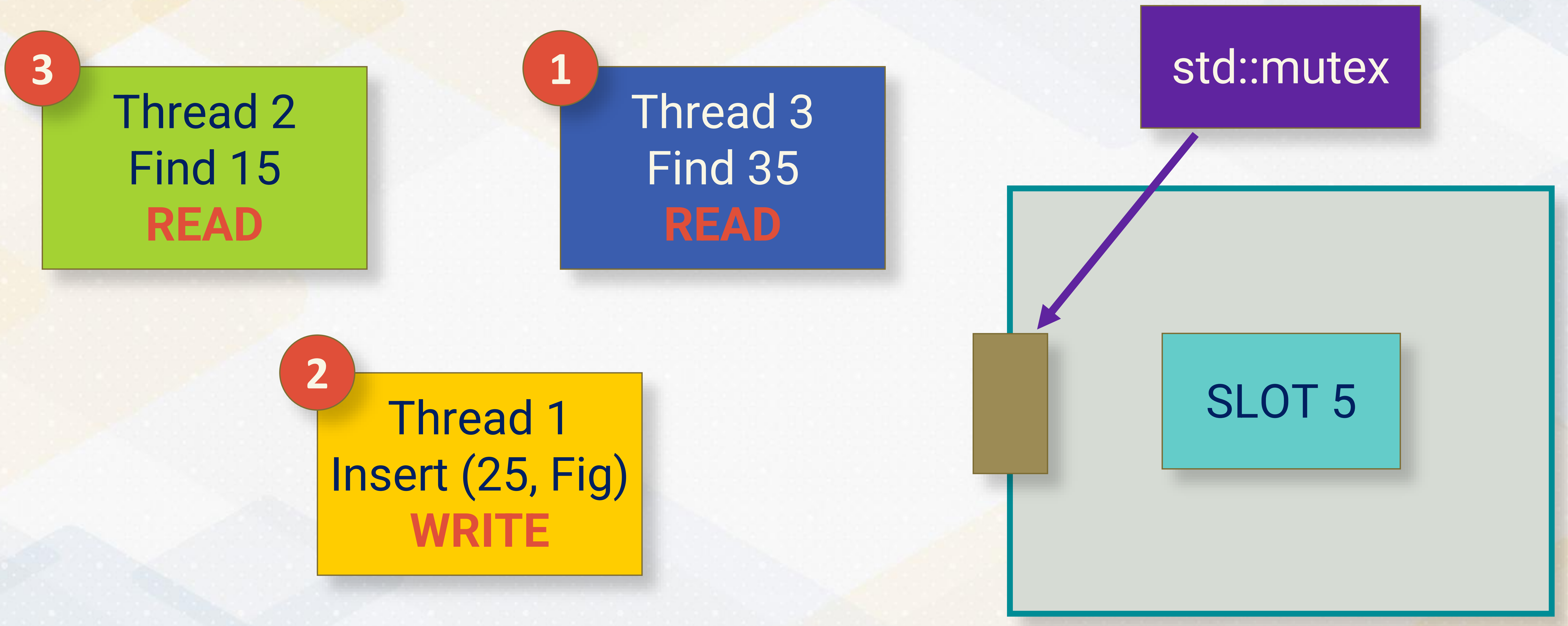

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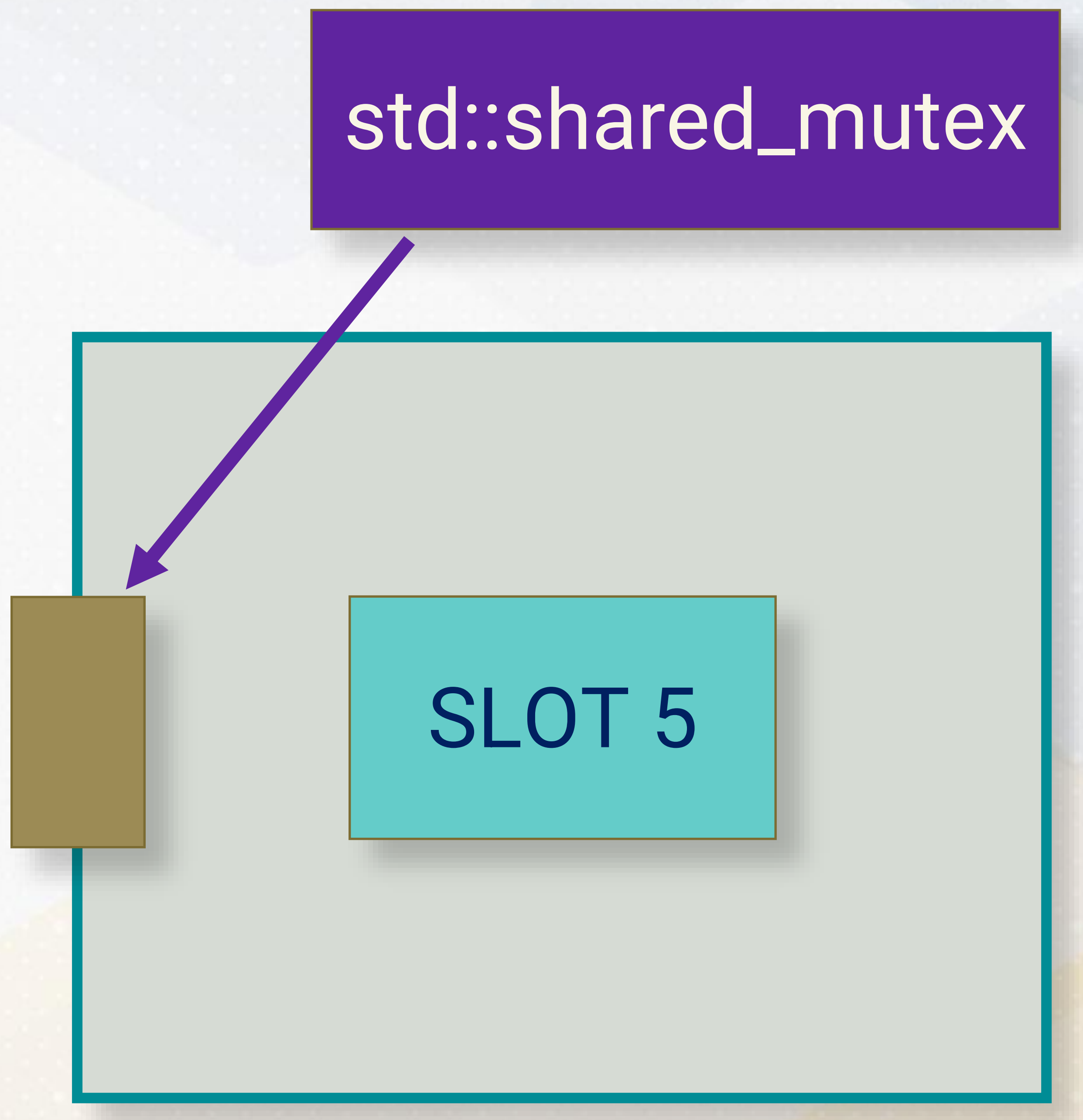

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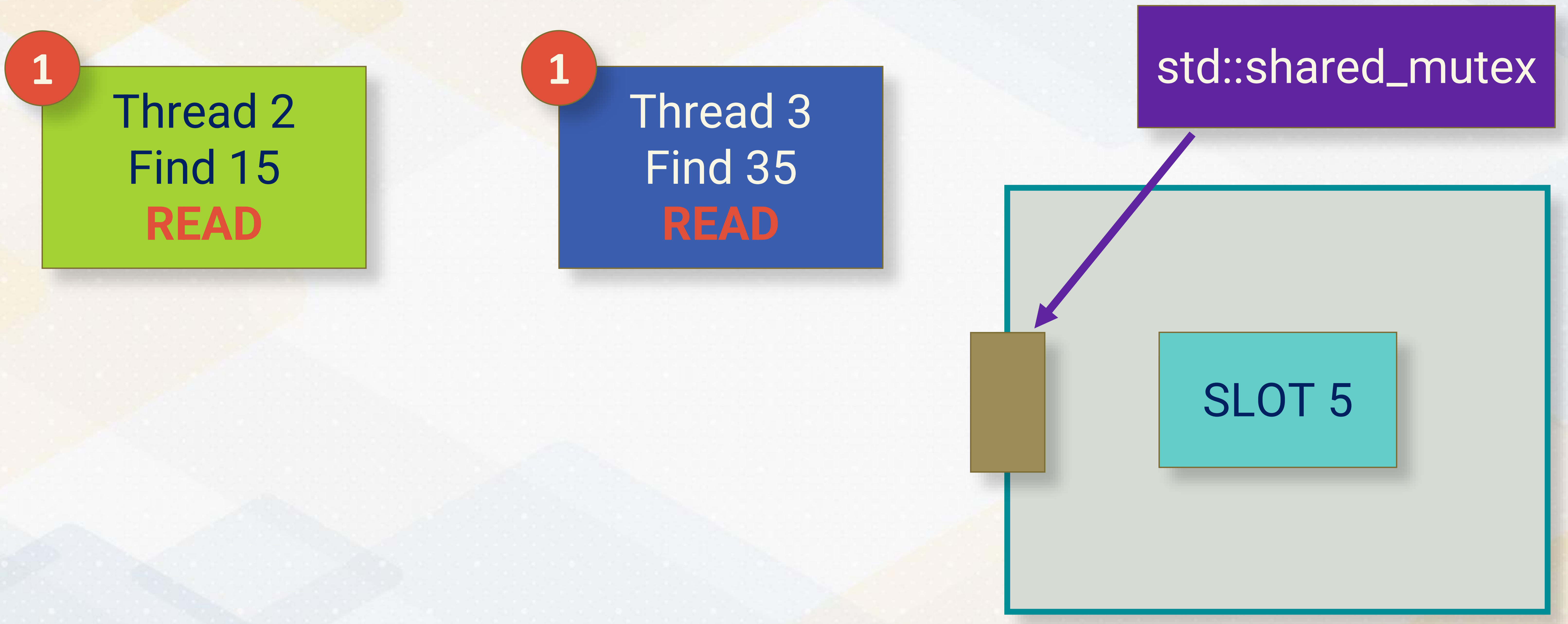

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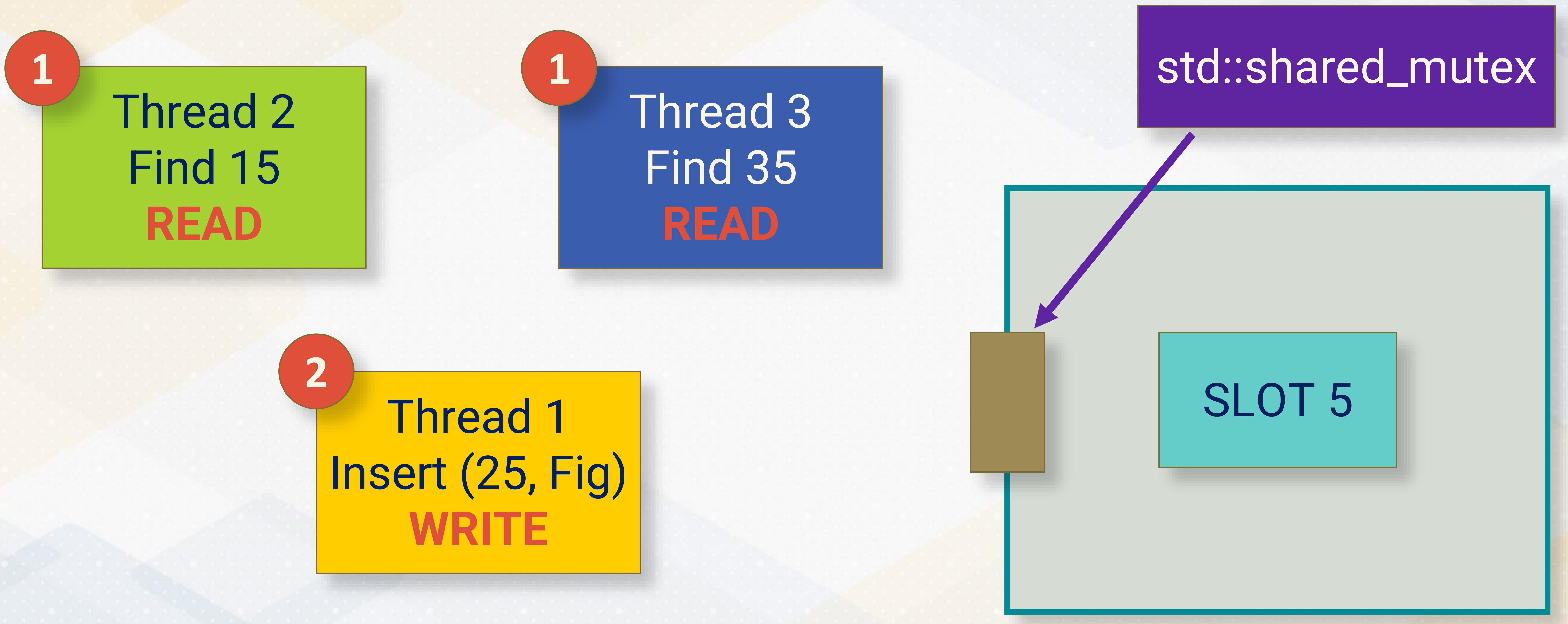


std::unique_lock<std::shared_mutex>

std::shared_mutex allows multiple threads to hold a read (shared) lock simultaneously while ensuring exclusive access for write operations.

```
mutable std::shared_mutex mutexes[capacity];
std::vector<std::unique_ptr<std::mutex>> mutexes;
 void insertOrUpdate(int key, int value) {
   size_t index = hashFunction(key); // Determine the slot
   do {
     // Exclusive lock for writing
     std::unique_lock<std::shared_mutex> lock(mutexes[index]);
     // Attempt to insert or update the slot
     if (conditions_met) {...}
   } while (not_inserted);
}
```

std::unique_lock<std::shared_mutex>

std::shared_lock<std::shared_mutex> allows multiple threads to read from the same slot concurrently, provided no thread is writing to it.

```
int getValue(int key) const {
    size_t index = hashFunction(key);
    size_t originalIndex = index;
    do {
      // Shared lock for reading
      std::shared_lock<std::shared_mutex> lock(mutexes[index]);
      // Check if the key is inside the slot or not
      // Calculate next slot index
    } while (index != originalIndex);
  }
}
```

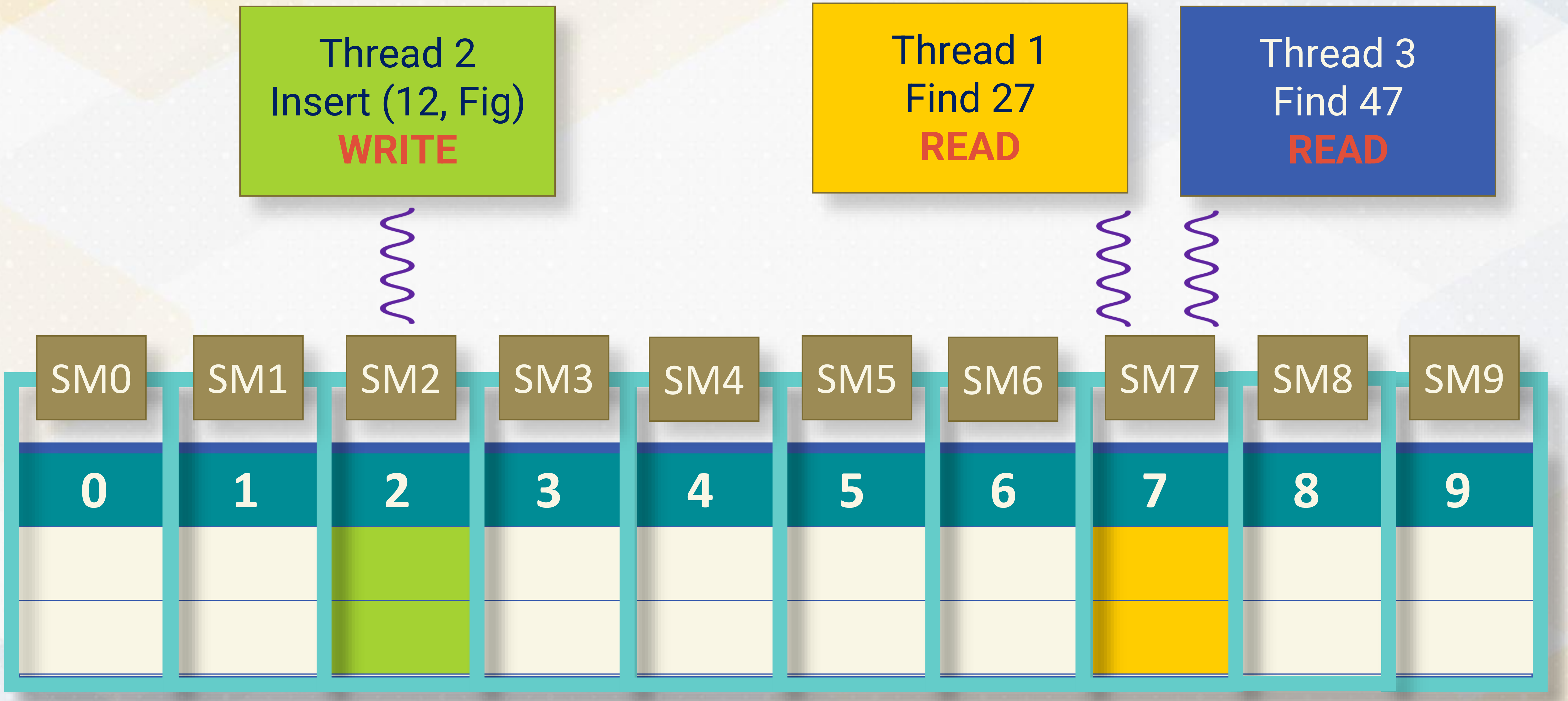
Exclusive Write Lock

Shared Resource	One Hash Table Slot
Mutex Type	std::unique_lock <std::shared_mutex>
Currently Accessing Threads	Thread 2 (**WRITE**)
Waiting Threads	Thread 1 (**READ**) Thread 3 (**READ**) …

Shared Read Lock

Shared Resource	One Hash Table Slot
Mutex Type	std::shared_lock <std::shared_mutex>
Currently Accessing Threads	Thread 1 (**READ**) Thread 3 (**READ**)
Waiting Threads	Thread 2 (**WRITE**) …

Fine-Grained Locking with Shared Mutex



Shared Mutex

Comparison of Protocols

Simulation Framework

Simulation Framework

Comparison of Locking Protocols

Need for Simulation

REAL THREADS & MUTEXES	SIMULATED THREADS & MUTEXES
Too fast for qualitative analysis	Controlled environment
Non-deterministic	Deterministic
Real deal	Only an approximation

Simulated Mutex

Allows only one thread to access the shared resource at a time.

```
def try_acquire(self):
    if not self.is_locked:
        self.is_locked = True
        return 'acquired’
    return 'waiting’
```

Simulated Shared Mutex

Allows multiple readers or a single writer.

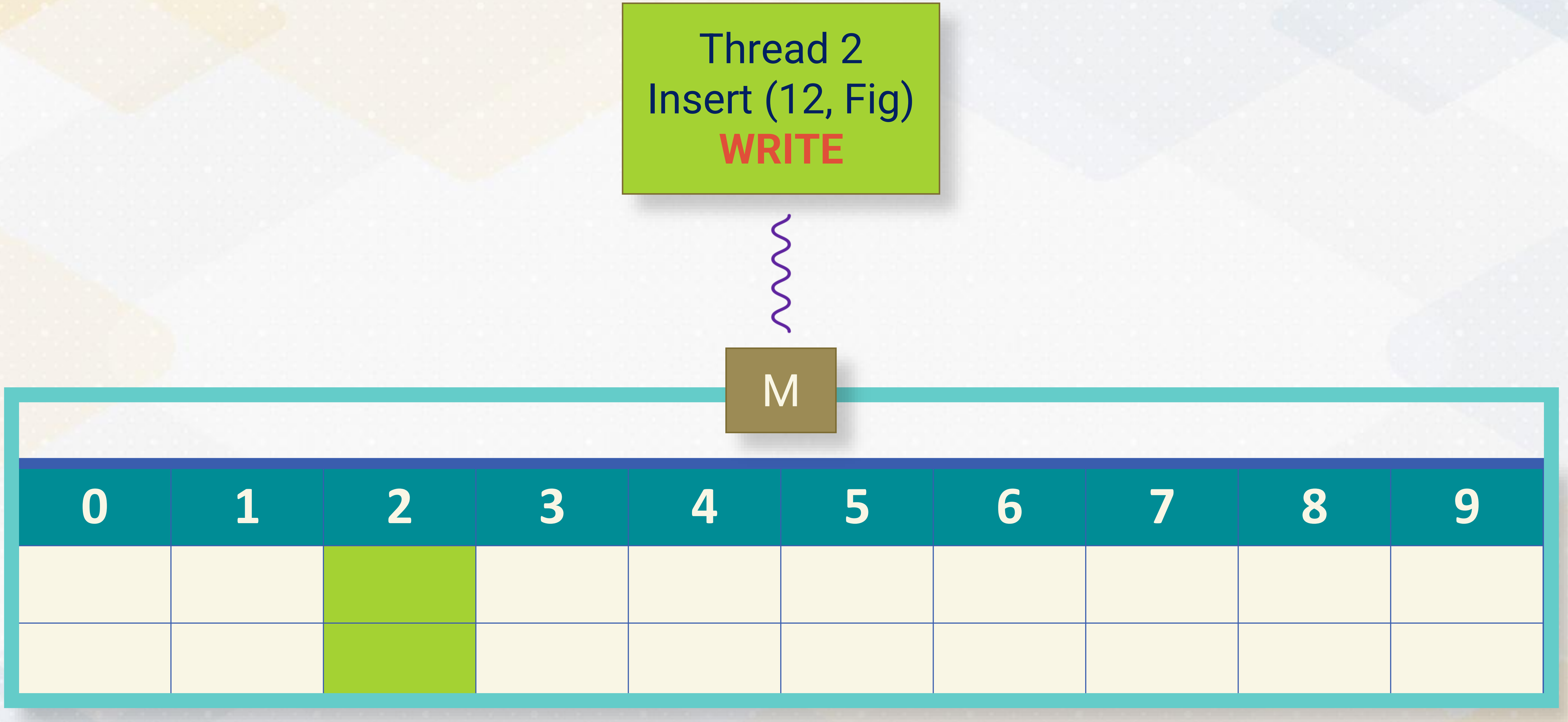
try_acquire_read	Grants read access unless a write lock is held
try_acquire_write	Grants write access if no reads or writes are active
release_read	Releases read lock
release_write	Releases write lock

Concurrent Hash Table

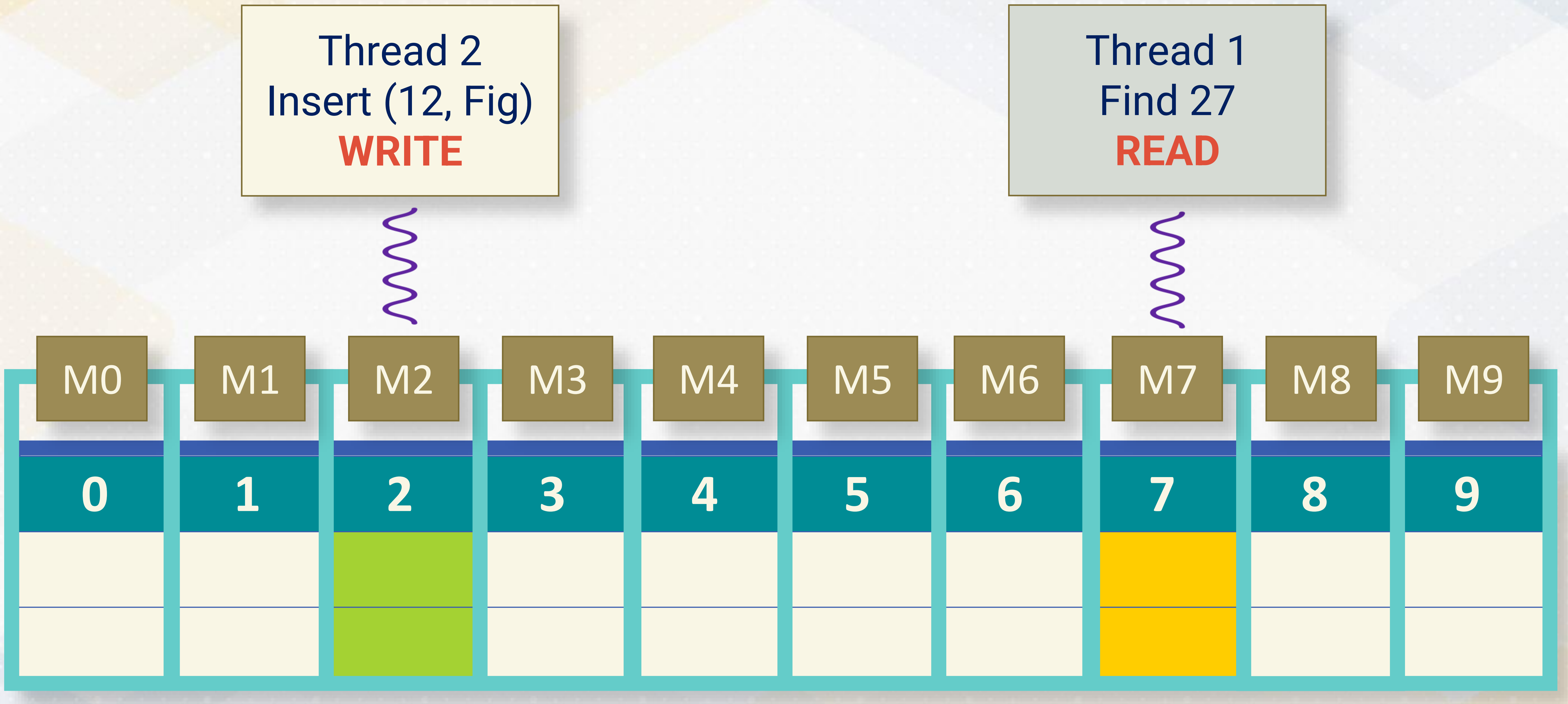
Models a hash table with three concurrency control protocols:

GLOBAL MUTEX	One mutex for entire table
MUTEX_PER_SLOT	One mutex for each slot
SHARED_MUTEX_PER_SLOT	One shared mutex for each slot

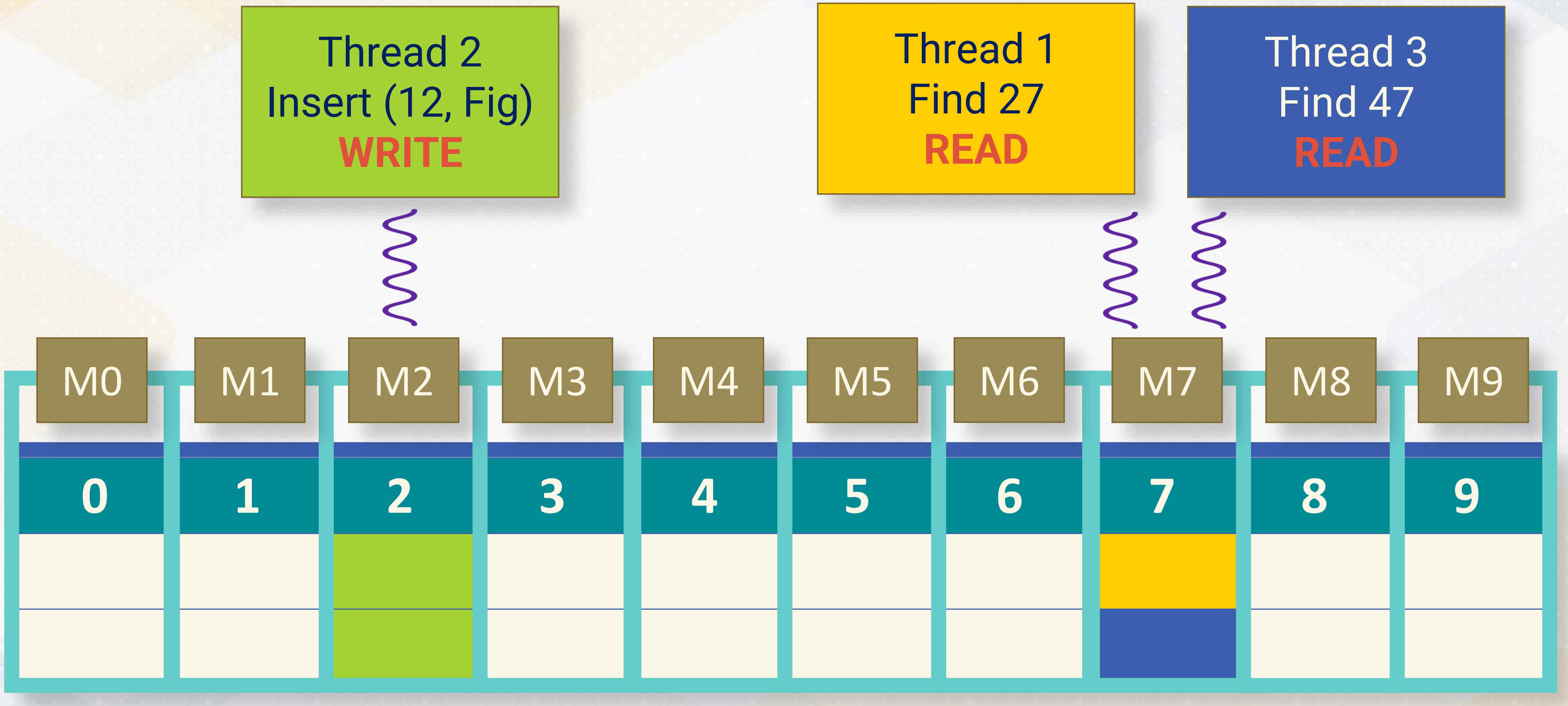
Protocol #1: global_mutex



Protocol #2: mutex_per_slot



Protocol #3: shared_mutex_per_slot



SimulatedThread

Simulates a thread performing a series of operations on the hash table. Step function manages the thread's operations based on its current state.

State	Description
Lock	Acquire lock
Find	Read to current slot
Insert	Write to current slot
Unlock	Release lock
Unlock_and_relock	Release lock and move to next slot

Thread-Safe Hash Table Simulator

Each thread can take only one step in a logical time step, and a thread can acquire a lock only if it was released in prior time step.

Time Step: 10
Thread-0: Attempt to lock for insert on slot 2 - waiting
Thread-1: Key 25 found at slot 5
Thread-2: Released lock on slot 3 – released

Hash Table Operation Trace

Each thread performs a sequence of operations:

Thread-0: insert (25, v25), find 25, find 15
Thread-1: find 35, insert (35, v35), find 25
Thread-2: find 45, find 25

Logical Time Step Count

Comparing the total logical time steps taken by different concurrency control protocols reveals insights into their efficiency for various operation traces.

GLOBAL MUTEX	30 steps
MUTEX_PER_SLOT	21 steps
SHARED_MUTEX_PER_SLOT	15 steps

Global Mutex

Time Step: 15

Thread-1: Released lock on table - released

Thread-2: Attempt to lock table - waiting

Time Step: 16

Thread-1: Attempt to lock table - acquired

Thread-2: Attempt to lock table - waiting

Mutex Per Slot

Time Step: 7

Thread-0: Attempt to lock slot 6 - acquired

Thread-1: Attempt to lock slot 5 – acquired

Thread-2: Attempt to lock slot 5 - waiting

Time Step: 8

Thread-0: Key 15 not found.

Thread-1: Found another key 25 at slot 5

Thread-2: Attempt to lock slot 5 - waiting

Shared Mutex Per Slot

Time Step: 7

Thread-0: Attempt to lock for find on slot 6 – acquired_read

Thread-1: Attempt to lock for insert on slot 5 – acquired_write

Thread-2: Attempt to lock for find on slot 6 – acuired_read

Time Step: 8

Thread-0: Key 15 not found.

Thread-1: Found another key 25 at slot 5

Thread-2: Key 45 not found.

Simulation Framework

Simulation Framework

Comparison of Locking Protocols